MIL-R-55348A(ER)
17 September 1980
SUPERSEDING
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2 February 1972

MILITARY SPECIFICATION

RADAR SET AN/PPS-5B
(UNITS OF)

This specification is approved for use by Electronics Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a lightweight combat surveillance radar system capable of detecting and locating both fixed and moving targets in the presence of ground clutter.

1.2 Units. This specification covers the requirements for the major units of Radar Set AN/PPS-5B. (see table II and 6.5).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The effective issue or revision of the following documents shall be that listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, unless (i) specific issues are set forth therefor in the cited specifications or (ii) issues different than those specified in the cited specifications are set forth in the solicitation. The date of the applicable DODISS and supplements thereto shall be as specified in the solicitation or contract.

SPECIFICATIONS

MILITARY

MIL-P-116	-	Preservation Packing, Methods of.
MIL-C-3885	-	Cable Assemblies and Cord Assemblies, Electrical.
MIL-M-13231	-	Marking of Electronic Items.
MIL-F-14072	-	Finishes for Ground Electronic Equipment.
MIL-M-49093	-	Handset H-251( )/U.

Beneficial comments (recommendations, additions, deletions and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Electronics Research and Development Command, Combat Surveillance and Target Acquisition Laboratory, ATTN: DELCS-PE, Fort Monmouth, N.J. 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5840

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-252	-	Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired, and Other Devices.
MIL-STD-454	-	Standard General Requirements for Electronic Equipment.
MIL-STD-461A	-	Electromagnetic Interference Characteristics, Requirements for Equipment.
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-810B	-	Environmental Test Methods.

DRAWINGS

DL-SM-A-610075	-	Radar Set AN/PPS-5( )
DL-SM-A-505770	-	Adapter Mount, Radar Set
GL-SM-B-323437	-	Mechanical Gages for Radar Set AN/PPS-5A
SC-B-61578	-	Grounding to Chassis
SC-A-46691	-	RF Cable Assembly Tests
SM-A-867587	-	Specification, EMC AN/PPS-5B
SM-A-867604		Reliability Test Specification
SPI1G00065	-	Special Packaging Instruction

OTHER PUBLICATIONS

Code of Federal Regulations

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. The contractor shall furnish first article samples for inspection and approval, if required by the invitation for bids and the contract. (see 4.3 and 6.2e).

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3.2 Construction. The equipment shall be constructed in accordance with the requirements of this specification and of the following drawings and documents:

DL-SM-A-610075 - Radar Set AN/PPS-5( ) includes the following:

DL-SM-A-686700	- Receiver-Transmitter, Radar, RT-965B/PPS-5A includes:
DL-SM-A-610106	- Drive, Antenna AB-992B/PPS-5 Assembly w/cover.
DL-SM-A-610077	- Reflector, Antenna, AS-2023/PPS-5.
DL-SM-A-686307	- Horn, Waveguide, AS-2413/PPS-5A.
DL-SM-A-610116	- Control-Indicator, Radar, C-4610B/PPS-5 includes:
DL-SM-A-610298	- Viewing Hood Assembly.
DL-SM-A-610121	- Stand, Control Radar.
DL-SM-A-610119	- Control Enclosure Assembly (cover).
DL-SM-A-610274	- Column Assembly, Tripod MX-7565B/PPS-5 includes:
DL-SM-A-610080	- Tripod, Radar, MT-2958A/PPS-5.
SM-D-686377	- Retainer Assembly/storage bracket
SM-B-540285	- Pin, Anchor
DL-SM-A-610086	- Boresight Assembly, includes:
SM-B-610371	- Mount, Telescope.
SM-C-610368	- Telescope
DL-SM-A-610089	- Battery Assembly BB-643A/PPS-5 includes:
DL-SM-A-610088	- Battery Box CY-3871A/PPS-5
DL-SM-A-610123	- Strap, Carrying, Battery Box
DL-SM-A-610078	- Power Supply, PP-4450C/PPS-5
DL-SM-A-610076	- Case, Radar Set CY-3872A/PPS-5 includes:
SM-B-545012	- Desiccant.
DL-SM-A-610081	- Harness, Control-Indicator, Radar
DL-SM-A-610082	- Harness, R/T Radar
DL-SM-A-610083	- Harness, Reflector
DL-SM-B-505770	- Adapter Mount, Radar Set.
DL-SM-A-610373	- Cable Assembly, Special, Electrical CS-12004B/PPS-5.
DL-SM-A-686897	- Cable Assembly, Special, Electrical CX-12068/PPS-5
MIL-H-49093	- Head Set H-251( )/U
DL-SM-A-867847	- Cable 50 ft.

3.3 Parts, materials and processes.

3.3.1 Cable assemblies. Cable assemblies shall be constructed in accordance with the applicable drawings and shall meet all the requirements of the following documents:

Specification	MIL-C-3885 (audio & power)
Drawing	SC-A-46691 (rf)

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3.3.2 Serial numbers. Each unit of Radar Set AN/PPS-5B shall be serial numbered.

3.3.3 Cleaning. In addition to Requirement 9 of MIL-STD-454, the following shall apply:

After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly.

3.3.4 Finish. The equipment shall be finished in accordance with MIL-F-14072 and the equipment drawings.

3.3.5 Marking. Marking shall be in accordance with Requirement 67 of MIL-STD-454 and front panel marking shall be Group I as described in MIL-M-13231.

3.3.6 Connections. Bared ends of wire leads to be terminated in solder-type terminal lugs shall be tinned, silver-plated, or lead-alloy coated. Electrical connections shall not be made by clamping between a metallic and a nonmetallic material. Welded electrical connections shall not be used where it may be necessary to disconnect and reconnect the circuit during servicing.

3.3.7 Grounding. Ground connection to shields and to other mechanical parts, except the chassis or frame, shall not be made to complete electrical circuits but only to eliminate high potential alternating current points. Intermediate-frequency transformers, however, may be grounded through their cases if other means of grounding are impracticable. Grounding to chassis shall conform to SC-B-61578.

3.3.8 Wiring. Wiring shall be in accordance with Requirement 20 of MIL-STD-454.

3.3.9 Other requirements. In addition to the above, the following requirements of MIL-STD-454 shall also apply:

a. Fungus-Inert Materials, per Requirement 4.

b. Soldering, per Requirement 5.

c. Insulating Materials, Electrical, per Requirement 11.

d. Fastener Hardware, per Requirement 12.

e. Structual Welding, per Requirement 13.

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f. Printed Wiring, per Requirement 17.

g. Derating of Electronic Parts and Materials, per Requirement 18.

h. Terminations, per Requirement 19.

i. Castings, per Requirement 21.

j. Adhesives, per Requirement 23.

k. Fibrous Material, Organic, per Requirement 44.

l. Indicator Lights, per Requirement 50.

m. Brazing, per Requirement 59.

3.3.10 Recycled, virgin and reclaimed materials. It is required that, to the maximum extent possible, recovered or reclaimed materials, in lieu of virgin materials, shall be used without jeopardizing the intended use of the manufactured item.

3.4 Interchangeability. Interchangeability shall be established through the use of Government furnished gages as prescribed by the drawings and subsequent documents (see 4.7).

3.5 Service conditions. The service conditions shall be as specified in SM-A-686622 and MIL-STD-810B. Table I indicates the specific conditions applicable to the complete equipment and to the individual units, when procured separately.

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TABLE I. Service conditions

Requirement	Complete radar set 1/	RT-965B/PPS-5 R/T	AB-992B/PPS-5 ant. drive	C-4610B/PPS-5 C/I	AS-2023/PPS-5 ant. reflector	AS-2413/PPS-5A antenna	PP-4450C/PPS-5 Pwr. supply	MT-2958A/PPS-5 tripod	MX-7565B/PPS-5 tripod/column	CY-3872A/PPS-5 case	CY-3871/PPS-5 bat. box	Each indiv. sub-unit	Pintle mount U/W ANPPS-5
Salt fog	b	X	X	X	X	X	X	X	X				
Low temperature	X	X	X	X	X	X	X	X	X	X	X		
High temperature	X	X	X	X	X	X	X	X	X	X	X		
Rain	a	X	X	X			X						
Humidity	X	X	X	X	X	X	X	X	X	X	X		X
Altitude	j	X	X	X			X			X	X		
Immersion	c	X		X		X	X		X	X	X		
Vacuum	e	X		X			X			X	X		
Dust	a	X	X	X			X						
Vibration	f	X	X	X			X			X			
Bounce	b	X	X	X			X			X	X		
Drop	g	X	X	X			X			X	X		
Bench handling	h	X	X	X			X					X	
Fungus	i	X	X	X			X			X	X	X	
Temperature shock	f	X		X			X			X			
Wind	X	X	X										

1/ When the complete radar set is being procured it shall be set up as for normal operation, using interconnecting cable assemblies as required, and those units listed below subjected to the indicated conditions. (see 3.5)

a. RT-965, AB-992, C-4610, PP-4450, AS-2023, AS-2413, MT-2958, MX-7565.
b. RT-965, AB-992, C-4610, PP-4450, CY-3871, AS-2023, AS-2413, MT-2958, MX-7565.
c. RT-965, C-4610, AS-2413, PP-4450, MX-7565, CY-3872, CY-3871.
d. RT-965, AB-992, C-4610, PP-4450.
e. RT-965, C-4610, PP-4450, CY-3872, CY-3871, MX-7565.
f. Test in manpack configuration.
g. Test transport case, fully loaded.
h. RT-965, AB-992, C-4610, PP-4450.
i. RT-965, AB-992, C-4610, PP-4450, AS-2023, AS-2413, MT-2958, MX-7565, CY-3872, CY-3871.
j. Transport case fully loaded, less batteries and electrolyte kits.

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3.6 Burn-in. Each equipment shall meet the requirements herein, without subsequent processing, after subjection to the burn-in procedure of 4.6.

3.7 Electromagnetic compatibility. The equipment shall meet the emission and susceptibility requirements of MIL-STD-461A, Change Notice 4(EL), table VI, as specified in SM-A-867587 (see 4.8.2).

3.8 Radar system performance. The complete radar system shall meet the requirements specified in SM-A-867537 Test Specification for Final Test, Radar Set AN/PPS-5B (see 4.9 and 6.5).

3.9 Unit performance requirements. Each unit shall meet the following requirements, independent of any other unit of the system (see 6.5).

3.9.1 Receiver-Transmitter, Radar, RT-965B/PPS-5A. The equipment shall meet the requirements of Part 2 of SM-A-867537, Test Specification for Final Test, Radar Set AN/PPS-5B.

3.9.1.1 Receiver-transmitter subassembly units.

a. Local Oscillator, per SM-A-686711
b. IF Signal Preamplifier, per SM-A-686787
c. AFC Preamplifier, per SM-A-686791
d. IF Amplifier, per SM-A-686811
e. Boxcar & Audio Amplifier, per SM-A-686806
f. AFC Amplifier, per SM-A-686813
g. Gate Generator, per SM-A-686807
h. Modulator Assembly, per SM-A-867478
i. Power Converter Assembly, per SM-A-867505
j. Motor Control Assembly, per SM-A-686809

    (1) Printed Wiring Board #1, per SM-A-686340 (Procedure)
    (2) Printed Wiring Board #2, per SM-A-867459 (Procedure)

k. Interlock Panel Assembly, per SM-A-686797
l. Relay Control Assembly, per SM-A-686795
m. Range Drive Assembly, per SM-A-686796
n. T/R Assembly, per SM-A-867439
o. Modulator Component Assembly, per SM-A-686338

3.9.1.2 Drive, Antenna, AB-992B/PPS-5, per SM-A-867499.

a. Azimuth Drive, per SM-A-686793
b. Commutator Assembly, per SM-A-686792
c. Azimuth Counter Assembly, per SM-A-686794

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3.9.2 Control-Indicator, Radar, C-4610B/PPS-5, per Section 3 of SM-A-867537.

a. Range Counter Assembly, per SM-A-686790
b. Remote Azimuth Counter, per SM-A-686789
c. "A" Display Assembly, per SM-A-686938
d. "B" Display Assembly, per SM-A-686800
e. Range Gated Filter Master Assembly, per SM-A-686798

(1) Range Gated Filter Module, per SM-A-686788
(2) Range Gated Filter Subassembly, per SM-A-686786

f. Range Amplifier Assembly, per SM-A-686802
g. Azimuth Servo Amplifier Assembly, per SM-A-686804

(1) 400 Cycle Generator, per SM-A-867481 (Procedure)
(2) Servo Amplifier, per SM-A-867482 (Procedure)

h. Remote Power Converter Assembly, per SM-A-686801

3.9.3 Power Supply PP-4450C/PPS-5, per SM-B-867467.

3.9.4 Tripod, Radar, MT-2958A/PPS-5, per SM-A-686814.

3.9.5 Horn, Waveguide, AS-2413( )/PPS-5A, per SM-D-686207.

3.9.6 Reflector, Antenna, AS-2023( )/PPS-5, per SM-A-867842.

3.10 Reliability. The equipment shall have a specified MTBF of 400 hours when used under the following conditions:

a. Ambient temperature: 50°C (+5 or -0 degrees C)
122°F (+9 or -0 degrees F)

b. Environment: Fixed ground

3.11 Systems safety engineering.

3.11.1 Personnel hazards. Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirement 1, shall be made applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly process, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with this requirement has been achieved shall be through a visual inspection (see 4.13).

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3.11.2 Edge rounding, exposed. Exposed edges shall be rounded to a minimum radius of 0.04 inch (1 mm), and exposed corners to a minimum of 0.5 inch (13 mm).

3.11.3 Radioactive materials. Radioactive materials shall not be used (e.g. luminous dials/markings, electron tubes, surge arrestors and lenses).

3.12 Workmanship. Workmanship shall be in accordance with the drawings and the workmanship requirements of MIL-STD-454, Requirement 9, and the applicable portions of the following paragraphs herein:

a.	Cleaning	3.3.3
b.	Finish	3.3.4
c.	Marking	3.3.5
d.	Connections	3.3.6
e.	Grounding	3.3.7
f.	Wiring	3.3.8
g.	Other requirements of MIL-STD-454	3.3.9

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (does not include packaging). (see 4.3)

b. Inspection covered by subsidiary documents. (see 4.4)

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before packaging. (see 4.5)

(2) Quality conformance inspection of packaging. (see 4.14)

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4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish eleven (11) first article units of the complete radar set AN/PPS-5B.

4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4, burn-in specified in 4.6, and the inspections specified for group A, group B, group C and group D (see table II, III, IV and V, respectively). The inspection shall be performed in the following order: 4.4, 4.6, group A, and group B for all units: group C as specified in table IV, and group D. After completion of group C environmental tests, conforming units shall be reinspected and shall pass all group A inspections.

4.3.3 First article data. The First Article Test Plan and Test Report(s) shall be as required in the contract or purchase order.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

Item	Where required
Cable assemblies	3.3.1
Finish	3.3.4
Marking	3.3.5

4.5 Quality conformance inspection of equipment before packaging. Every equipment shall be subjected to the preconditioning burn-in procedure of 3.6 prior to conformance inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.6. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in table II. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and the AQL's indicated in table II.

4.5.1.1 Order of inspection within group A. Group A inspection shall be performed in an order satisfactory to the Government.

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TABLE II. Group A inspection

Inspection	Req para	Insp para	AQL 1/ Major	AQL 1/ Minor
Visual and mechanical				
R-T Unit RT-965B/PPS-5A	3.12	4.11	1.5%	6.5%
Remote Unit C-4610B/PPS-5	3.12	4.11	1.5%	6.5%
Power Supply PP-4450C/PPS-5	3.12	4.11	1.5%	6.5%
Tripod Assy MT-2958A/PPS-5	3.12	4.11	1.5%	6.5%
Column Assy MX-7565B/PPS-5	3.12	4.11	1.5%	6.5%
Antenna Drive AB-992B/PPS-5	3.12	4.11	1.5%	6.5%
Antenna AS-2412/PPS-5				
Horn,Waveguide AS-2413/PPS-5A	3.12	4.11	1.5%	6.5%
Reflector, Antenna, AS-2023/PPS-5	3.12	4.11	1.5%	6.5%
Telescope Assy	3.12	4.11	1.5%	6.5%
Battery Box CY-3871A/PPS-5	3.12	4.11	1.5%	6.5%
Transit Case CY-3872( )/PPS-5	3.12	4.11	1.5%	6.5%
Carrying Harnesses	3.12	4.11	1.5%	6.5%
Pintle Mount Assy	3.12	4.11	1.5%	6.5%
Stand, Control, Radar	3.12	4.11	1.5%	6.5%
Operational and electrical tests:				
System performance	3.8	4.9	1.5%	2/
Unit performance	3.9	4.10		
Receiver-Transmitter RT-965B /PPS-5A (complete unit)	3. 9.1	4.10.1	1.5%	2/
Local Oscillator	3. 9.1.1a	4.10.1.1a	1.5%	2/
I.F. Signal Preamplifier	3. 9.1.1b	4.10.1.1b	1.5%	2/
A.F.C. Preamplifier	3. 9.1.1c	4.10.1.1c	1.5%	2/
I.F. Amplifier	3. 9.1.1d	4.10.1.1d	1.5%	2/
Boxcar & Audio Amplifier	3. 9.1.1e	4.10.1.1e	1.5%	2/
A.F.C. Amplifier	3. 9.1.1f	4.10.1.1f	1.5%	2/
Gate Generator	3. 9.1.1g	4.10.1.1g	1.5%	2/
Modulator Assembly	3. 9.1.1h	4.10.1.1h	1.5%	2/
Power Converter Assembly	3. 9.1.1i	4.10.1.1i	1.5%	2/
Motor Control Assembly	3. 9.1.1j	4.10.1.1j	1.5%	2/
Printed Wiring Board #1	3. 9.1.1j(1)	4.10.1.1j(1)	1.5%	2/
Printed Wiring Board #2	3. 9.1.1j(2)	4.10.1.1j(2)	1.5%	2/

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TABLE II. Group A inspection - continued

Inspection	Req para	Insp para	AQL 1/ Major	AQL 1/ Minor
Interlock Panel Assembly	3.9.1.1k	4.10.1.1k	1.5%	2/
Relay Control Assembly	3.9.1.1l	4.10.1.1l	1.5%	2/
Range Drive Assembly	3.9.1.1m	4.10.1.1m	1.5%	2/
T/R Assembly	3.9.1.1n	4.10.1.1n	1.5%	2/
Modulator Component Assembly	3.9.1.1o	4.10.1.1o	1.5%	2/
Drive, Antenna AB-992B/PPS-5	3.9.1.2	4.10.1.2	1.5%	2/
Azimuth Drive	3.9.1.2a	4.10.1.2a	1.5%	2/
Commutator Assembly	3.9.1.2b	4.10.1.2b	1.5%	2/
Azimuth Counter Assembly	3.9.1.2c	4.10.1.2c	1.5%	2/
Control-Indicator, Radar C-4610B/PPS-5	3.9.2	4.10.2	1.5%	2/
Range Counter Assembly	3.9.2a	4.10.2a	1.5%	2/
Remote Azimuth Counter	3.9.2b	4.10.2b	1.5%	2/
"A" Display Assembly	3.9.2c	4.10.2c	1.5%	2/
"B" Display Assembly	3.9.2d	4.10.2d	1.5%	2/
Range Gated Filter Master Assembly	3.9.2e	4.10.2e	1.5%	2/
Range Gated Filter Module	3.9.2e(1)	4.10.2e(1)	1.5%	2/
Range Gated Filter Subassembly	3.9.2e(2)	4.10.2e(2)	1.5%	2/
Range Amplifier Assembly	3.9.2f	4.10.2f	1.5%	2/
Azimuth Servo Amplifier Assembly	3.9.2g	4.10.2g	1.5%	2/
400 Cycle Generator	3.9.2g(1)	4.10.2g(1)	1.5%	2/
Servo Amplifier	3.9.2g(2)	4.10.2g(2)	1.5%	2/
Remote Power Converter Assembly	3.9.2h	4.10.2h	1.5%	2/
Power Supply PP-4450C/PPS-5	3.9.3	4.10.3	1.5%	2/
Tripod, Radar MT-2958A/PPS-5	3.9.4	4.10.4	1.5%	2/
Horn, Waveguide AS-2413/PPS-5A	3.9.5	4.10.5	1.5%	2/
Reflector, Antenna AS-2023/PPS-5	3.9.6	4.10.6	1.5%	2/

1/ The AQL's are applicable when individual items are procured separately.

2/ All defects shall be considered major.

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4.5.2 Group B inspection. Group B inspection shall normally be performed in inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. This inspection shall conform to table III and to the special inspection levels of table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The group B sampling plans, for the AQL's listed in table III, shall be inspection level S-4.

TABLE III. Group B inspection.

Inspection	Req. para.	Insp. para.	AQL
Interchangeability	3.4	4.7	6.5%

4.5.3 Group C inspection. This inspection shall consist of the tests specified in table IV and shall be performed on sample units that have been subjected to and met group A and group B inspection. Sample units shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for group C inspection. Three (3) sample units shall be selected at random for all inspections in table IV from every one hundred (100) or fraction thereof produced. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot (see 4.5.3).

4.5.3.2 Order of inspection within group C. Group C inspection shall be performed in accordance with table IV.

4.5.3.3 Group C failures. Actions required relative to group C failures shall be as specified in the contract (see 6.3).

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TABLE IV. Group C inspection.

Inspection	Req. para.	Test para.	Order of inspection Unit 1	Unit 2	Unit 3
Non-damaging					
Antenna patterns 3/			1	1	1
Temperature	3.5	4.8			
Low	3.5	4.8	2		
High	3.5	4.8	3		
Rain	3.5	4.8	5		
Altitude (elevation) 2/	3.5	4.8		2	
Leakage (immersion)	3.5	4.8			2,8
Dust	3.5	4.8			3
Electromagnetic compatibility	3.7	4.8.2		3	
Wind 4/	3.5	4.8	7		
Temp. shock	3.5	4.8	4		
Potentially damaging					
Humidity (moisture resistance)	3.5	4.8	6		
Fungus 1/	3.5	4.8			9
Salt fog 1/	3.5	4.8		4	
Bounce, including resonance search or fatigue-type vibration	3.5	4.8			4
Shock, bench handling	3.5	4.8			5
Shock, drop	3.5	4.8			6
Vibration	3.5	4.8			7

1/ The equipment shall be thoroughly washed, cleaned, dried, and refurbished, as specified in contract, before proceeding with subsequent tests.

2/ The immersion test shall be run both before and after the potentially damaging test series.

3/ The antenna pattern test shall be conducted per SM-A-686830, on the receiver-transmitter unit in lieu of the test fixture reference in the test drawing.

4/ The wind test shall be performed on the designated test units in the first articles only. It shall not be repeated in production group C tests. The Government Wind Test Facility at Fort Huachuca will be made available (Contractor supplies his own operators for the radar) if no other suitable facilities are available to the contractor.

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4.5.4 Reinspection of conforming group B and group C sample units. Unless otherwise specified, sample units which have been subjected to and passed both group B and group C inspection may be accepted on contract or order provided all damage is repaired and the sample units are resubjected to and pass group A inspection.

4.5.5 Group D inspection. This inspection shall consist of the tests specified in table V and shall be performed on units from lots which have been subjected to and met group A and group B inspections.

4.5.5.1 Sampling for group D inspection. Eight (8) samples shall be selected at random from every 150 units or fraction thereof produced.

4.5.5.2 Group D failures. Action required relative to group D failures shall be as specified in the contract or purchase order.

TABLE V. Group D inspection.

Inspection	Req. para.	Insp. para.
Reliability	3.10	4.12

4.5.6 Reinspection of conforming group D sample units. Unless otherwise specified, sample units which have been subjected to and passed group D inspection may be accepted on the contract or purchase order provided all visible and known damage is repaired and the sample units are resubjected to and pass group A inspection.

4.6 Burn-in. Each equipment shall be subjected to the burn-in procedure of SM-A-867529 prior to subjection to group A testing. The System Alignment Procedure SM-A-867525 shall be performed prior to subjection to burn-in test.

4.7 Inspection for interchangeability. This inspection shall be accomplished with the Government furnished gages listed in GL-SM-B-323437. Gaging may be performed at any appropriate point in the manufacturing cycle. If a part/item/assembly being checked by a gage does not pass, an alternate manual inspection of the part/item/assembly shall be performed for compliance. Verification of drawing requirements takes precedence over gage acceptance.

4.8 Service condition tests. Service condition tests shall be performed as detailed in SM-A-686622. Each unit subjected to these tests shall have passed group A and group B tests and shall be resubmitted to, and pass these group A tests after completion of all service condition testing.

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4.8.1 Testing. Where testing is specified during service condition cycling the conditions specified in SM-A-686944 shall be met. These same conditions shall be met at the conclusion of each separate service condition test.

4.8.2 Electromagnetic compatibility. The equipment shall be tested for compliance with the requirements of 3.7 using the approved test methods of MIL-STD-462, Notice 3(EL) as modified by SM-A-867587.

4.9 System tests. System testing per SM-A-867523, Final Test Procedure, shall be performed to determine compliance with the requirements of 3.8 (See 4.10.6).

4.10 Unit tests. Each unit shall be tested as specified in the following paragraphs. These tests shall be conducted independently of any other unit of the radar set.

4.10.1 Receiver-Transmitter, Radar RT-965B/PPS-5A. This unit shall be tested as specified in SM-A-867535, Alignment Procedure for R/T Radar.

4.10.1.1 Receiver-Transmitter subassembly units.

a. Local Oscillator, per SM-A-686848
b. IF Signal Preamplifier, per SM-A-867541
c. AFC Preamplifier, per SM-A-686640
d. I.F. Amplifier, per SM-A-867501
e. Boxcar & Audio Amplifier, per SM-A-686644
f. AFC Amplifier, per SM-A-686652
g. Gate Generator, per SM-A-686646
h. Modulator Assembly, per SM-A-867475
i. Power Converter Assembly, per SM-A-867378
j. Motor Control Assembly, per SM-A-867461

(1) Printed Wiring Board #1, per SM-A-686340
(2) Printed Wiring Board #2, per SM-A-867459

k. Interlock Panel Assembly, per SM-A-867451
l. Relay Control Assembly, per SM-A-686658
m. Range Drive Assembly, per SM-A-686680
n. T/R Assembly, per SM-A-867440
o. Modulator Component Assembly, per SM-A-867476

4.10.1.2 Drive, Antenna, AB-992B/PPS-5, per SM-A-867468.

a. Azimuth Drive, per SM-A-686686
b. Commutator Assembly, per SM-A-686684
c. Azimuth Counter Assembly, per SM-A-686688

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4.10.2 Control-Indicator, Radar, C-4610B/PPS-5, per SM-A-867545 .

a. Range Counter Assembly, per SM-A-686701
b. Remote Azimuth Counter, per SM-A-686674
c. "A" Display Assembly, per SM-A-686938 and SM-A-609996
d. "B" Display Assembly, per SM-A-686662 and SM-A-609996
e. Range Gated Filter Master Assembly, per SM-A-867388

(1) Range Gated Filter Module, per SM-A-686671
(2) Range Gated Filter Subassembly, per SM-A-686670

f. Range Amplifier Assembly, per SM-A-686676
g. Azimuth Servo Amplifier Assembly, per SM-A-867484

(1) 400 Cycle Generator, Alignment Procedure, per SM-A-867481
(2) Servo Amplifier, Alignment Procedure, per SM-A-867482

h. Remote Power Converter Assembly, per SM-A-867486

4.10.3 Power Supply PP-4450C/PPS-5, per SM-A-867464 .

4.10.4 Tripod, Radar, MT-2958A/PPS-5, per SM-A-686703 .

4.10.5 Horn, Waveguide, AS-2413/PPS-5A, per SM-D-686207 .

4.10.6 Reflector, Antenna, AS-2023/PPS-5, per SM-A-867843 .

4.11 Visual and mechanical inspection. The equipment shall be examined for the defects listed in MIL-STD-252.

4.12 Reliability acceptance tests. The contractor shall conduct the reliability acceptance tests in accordance with the Reliability Test Specification SM-A-867604 (see 3.10).

4.13 Systems safety inspection. An inspection shall be performed to verify compliance with those portions of 3.11 which can be determined visually.

4.13.1 Lens test, thorium content. A test shall be performed to determine that the thorium content of any lens covered in Title 10, Code of Federal Regulations, Part 40.13(c) is below 500 parts/million by weight. Results of this test shall be included in the Ionizing Radiation Test Report if required in the contract.

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4.14 Quality conformance inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of section 5.

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with SPI1G00065.

6. NOTES

6.1 Intended use. Radar Set AN/PPS-5B is a non-coherent pulse doppler type which uses ground clutter in the vicinity of the target as a reference in order to obtain a doppler frequency modulated target return. The equipment detects fixed targets and moving targets with radial velocities between 1 mph and 35 mph. It provides a choice of automatic operation modes for area surveillance, with various selectable azimuth sector widths, out to ranges of 10,000 meters for vehicles and 5,000 meters for individual personnel. The radar set operates over frequency range of 16.0 to 16.5 GHz (Ku band) and, with receiver/transmitter unit only, provides aural indication of target detection out to 5000 meters range and, at the option of operator, a searchlighting or azimuthal scanning operational mode, the latter being either manual or automatic. The addition of a remote unit (remote control-indicator), which can be located up to 50 feet away from the basic radar, simultaneously provides both range-amplitude display, and range/azimuth display on two separate cathode-ray tube screens to a maximum range of 10,000 meters. An aural output is available at the remote unit. The equipment is so constructed as to be man-transportable by no more than three men, transportable by vehicle, and operable by one man.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Complete equipment or individual units to be procured.

c. Levels of preservation and packaging and packing.

d. When rough handling and functional tests are required.

e. Number of first articles to be submitted for approval (see 3.1).

f. Marking and shipping of samples.

g. Place of final inspection.

h. Technical literature required.

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6.3 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action (see 4.5.3.3).

6.4 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.5 Cross reference of requirements and test procedure for units of Radar Set AN/PPS-5B.

	Requirement drawing	Test drawing
Radar Set AN/PPS-5B	SM-A-867537	SM-A-867523
Receiver-Transmitter, Radar RT-965B/PPS-5A	SM-A-867537	SM-A-867535
Local Oscillator	SM-A-686711	SM-A-686848
IF Signal Preamplifier	SM-A-686787	SM-A-867541
AFC Preamplifier	SM-A-686791	SM-A-686640
IF Amplifier	SM-A-686811	SM-A-867501
Boxcar & Audio Amplifier	SM-A-686806	SM-A-686644
AFC Amplifier	SM-A-686813	SM-A-686652
Gate Generator	SM-A-686807	SM-A-686646
Modulator Assembly	SM-A-867478	SM-A-867475
Power Converter Assembly	SM-A-867505	SM-A-867378
Motor Control Assembly	SM-A-686809	SM-A-867461
(1) Printed Wiring Board #1	SM-A-686340	SM-A-686340
(2) Printed Wiring Board #2	SM-A-867459	SM-A-867459
Interlock Panel Assembly	SM-A-686797	SM-A-867451
Relay Control Assembly	SM-A-686795	SM-A-686658
Range Drive Assembly	SM-A-686796	SM-A-686680
T/R Assembly	SM-A-867439	SM-A-867440
Modulator Component Assembly	SM-A-686338	SM-A-867476
Drive Antenna, AB-992B/PPS-5	SM-A-867499	SM-A-867468
Azimuth Drive	SM-A-686793	SM-A-686686
Commutator Assembly	SM-A-686792	SM-A-686684
Azimuth Counter Assembly	SM-A-686794	SM-A-686688

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	Requirement drawing	Test drawing
Control-Indicator, Radar C-4610B/PPS-5	Section 3 of SM-A-867537	SM-A-867545
Range Counter Assembly	SM-A-686790	SM-A-686701
Remote Azimuth Counter	SM-A-686789	SM-A-686674
"A" Display Assembly	SM-A-686938	SM-A-686938 and SM-A-609996
"B" Display	SM-A-686800	SM-A-686662 and SM-A-609996
Range Gated Filter Master Assy	SM-A-686798	SM-A-867388
(1) Range Gated Filter Module	SM-A-686788	SM-A-686671
(2) Range Gated Filter Subassy	SM-A-686786	SM-A-686670
Range Amplifier Assembly	SM-A-686802	SM-A-686676
Azimuth Servo Amp Assy	SM-A-686804	SM-A-867484
(1) 400 Cycle Generator	SM-A-867481	SM-A-867481
(2) Servo Amplifier	SM-A-867482	SM-A-867482
Remote Power Converter Assy	SM-A-686801	SM-A-867486
Power Supply PP-4450C/PPS-5	SM-B-867467	SM-A-867464
Tripod, Radar MT-2958A/PPS-5	SM-A-686814	SM-A-686703
Horn, Waveguide AS-2413/PPS-5A	SM-D-686207	SM-D-686207
Reflector, Antenna AS-2023/PPS-5	SM-A-867842	SM-A-867843

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MIL-R-55348A(ER) RADAR SET AN/PPS-5B (UNITS OF)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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